

UML and System Diagrams: Types and Purposes

A practical reference for choosing diagrams to document a system, including examples for a Corporate Policy AI Assistant.

Common Diagram Types

Diagram	Primary Purpose	What It Shows	Corporate Policy AI Assistant Example
Use Case Diagram	Shows what the system does and who interacts with it.	Actors, use cases, system boundary, relationships.	Employee -> Ask Policy Question -> Corporate Policy AI Assistant
Activity Diagram	Shows the workflow or business process from beginning to end.	Activities, decisions, branches, and parallel activities.	Ask question -> validate -> search policies -> generate answer -> display response
Sequence Diagram	Shows how components communicate over time.	Participants, messages, calls, responses, and execution order.	User -> React UI -> API -> Azure AI Search -> LLM -> API -> User
System Sequence Diagram	Shows high-level interactions between an external actor and the system as a whole.	Actor, system boundary, requests, and responses.	Employee -> Policy Assistant: Submit Question; Policy Assistant -> Employee: Return Answer
Class Diagram	Shows the static structure of software or information objects.	Classes, attributes, methods, and relationships.	Policy, PolicyChunk, UserQuery, SearchResult, ChatResponse
Component Diagram	Shows the major software/services and their relationships.	Applications, APIs, databases, services, and interfaces.	React UI -> Azure Function/API -> Azure AI Search -> Foundry LLM
Deployment Diagram	Shows where system components are physically or cloud deployed.	Servers, cloud services, containers, networks, and deployed applications.	Azure hosting React, Functions, AI Search, Blob Storage, and Microsoft Foundry
State Machine Diagram	Shows how an object or process changes state.	States, events, and transitions.	Query: Submitted -> Validated -> Searching -> Reasoning -> Completed/Failed
Data Flow Diagram (DFD)	Shows how data moves through the system.	Data sources, processes, data stores, and destinations.	Blob -> ingestion -> embeddings -> vector index; query -> search -> LLM -> response
Entity Relationship Diagram (ERD)	Shows the structure and relationships of stored data.	Entities, fields, keys, and relationships.	Policy -> Policy Section -> Policy Chunk -> Embedding
Communication Diagram	Shows how system objects/components collaborate.	Objects/components and numbered communications.	UI, API, Search Service, and LLM exchanging messages
Package Diagram	Organizes a large system into logical groups or modules.	Packages/modules and dependencies.	User Interface, API, RAG Services, AI Services, Data Services, Security
Context Diagram	Shows the system at the highest architectural level.	System and external users/systems.	Employees, Administrators, Policy Repository -> Corporate Policy AI Assistant
Architecture Diagram	Provides an overall technical view of the solution.	Applications, cloud services, networks, data stores, AI components, integrations.	React + API + Azure AI Search + Blob Storage + Foundry Embeddings + Foundry LLM

Which Diagram Should You Use?

Question You Want to Answer	Best Diagram
Who uses the system and what can they do?	Use Case Diagram
What steps happen in the business/process flow?	Activity Diagram
In what order do the technical components communicate?	Sequence Diagram
How does data move?	Data Flow Diagram
What applications/services make up the solution?	Component Diagram
Where are those components deployed?	Deployment Diagram
How is the data structured?	Entity Relationship Diagram (ERD)
What is the overall system and what interacts with it?	Context Diagram
What is the complete technical architecture?	Architecture Diagram

Recommended Set for the Corporate Policy AI Assistant

For the Corporate Policy AI Assistant, the most useful core set is: **Context Diagram, Use Case Diagram, Activity Diagram, Sequence Diagram, Data Flow Diagram, and Architecture/Component Diagram**. Together, these describe the solution from the business-user perspective through the technical RAG/AI implementation.